

Geant4 Tutorial Course

Jefferson Lab, Newport News, VA, USA

August 18-22 2025

Electromagnetic Physics I

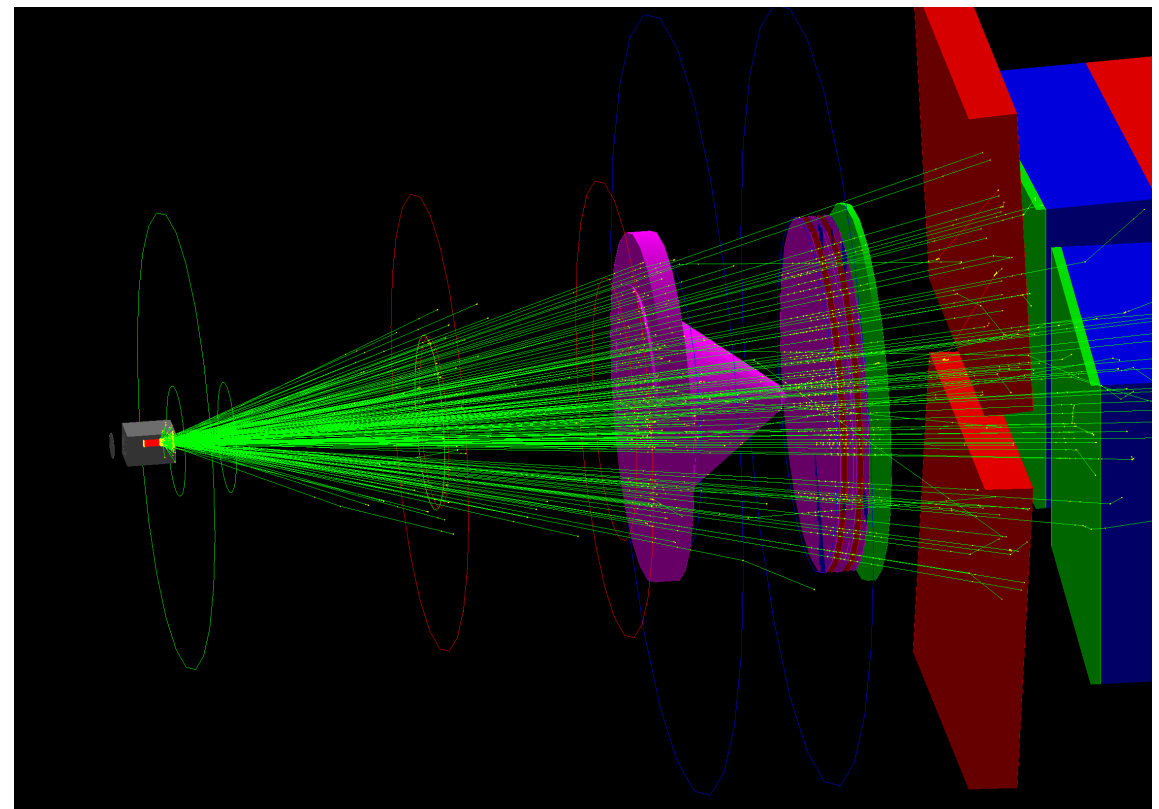
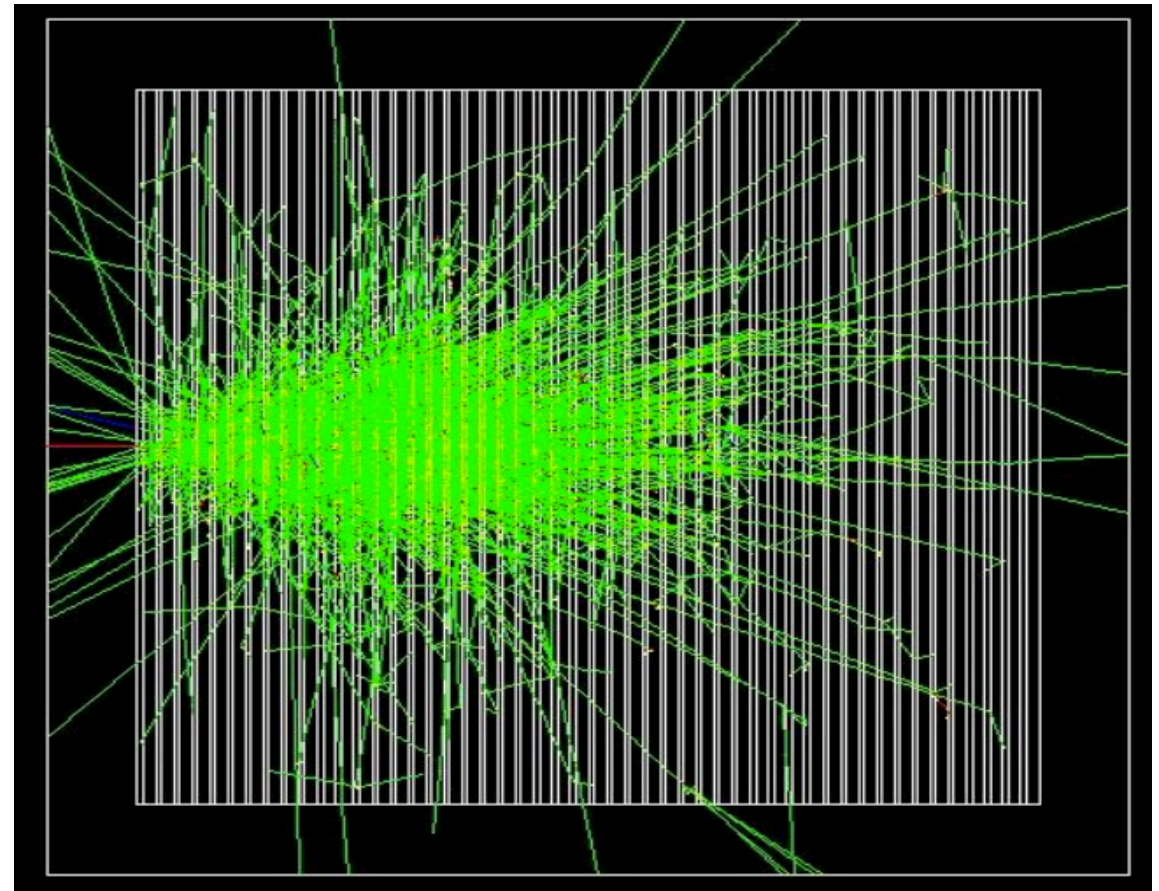
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Content from Mihaly Novak (CERN, EP-SFT) and Vladimir Ivantchenko (Tomsk State University)

Introduction

- ◆ Electromagnetic (EM) physics overview
- ◆ Gamma processes
- ◆ Charged particles processes
- ◆ Production Threshold
- ◆ EM Physics Constructors
- ◆ User Interfaces to EM physics

HEP Calorimeter



Medical Linac

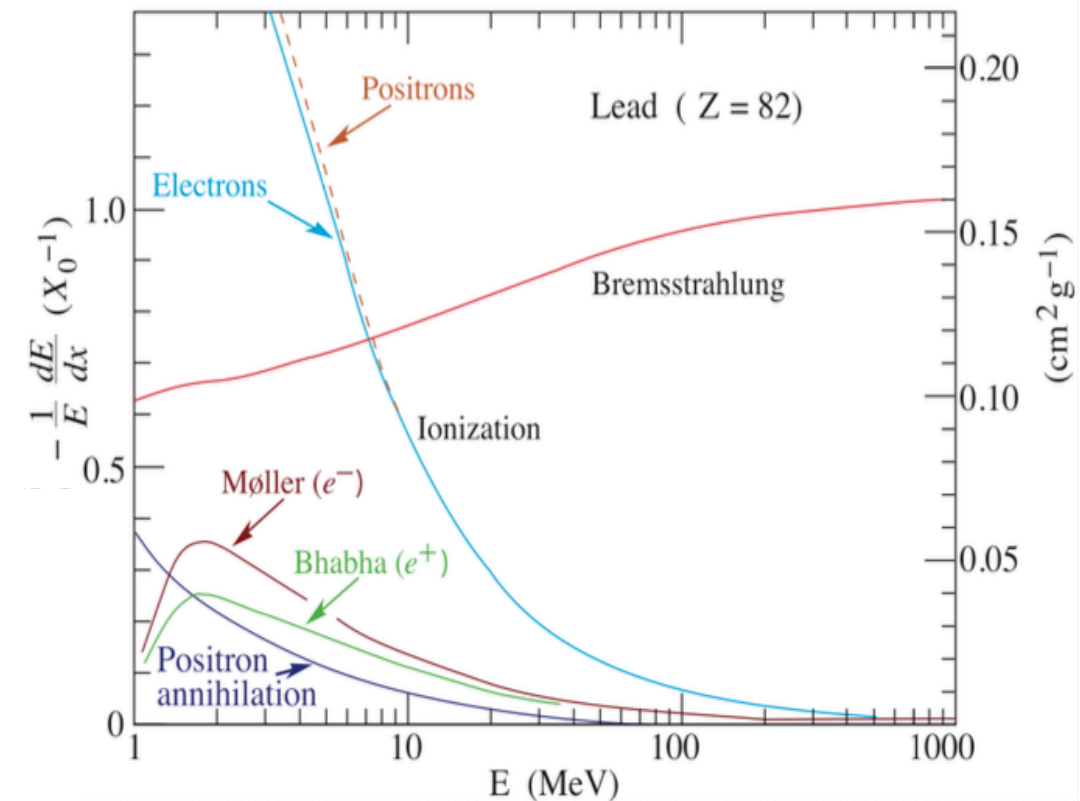
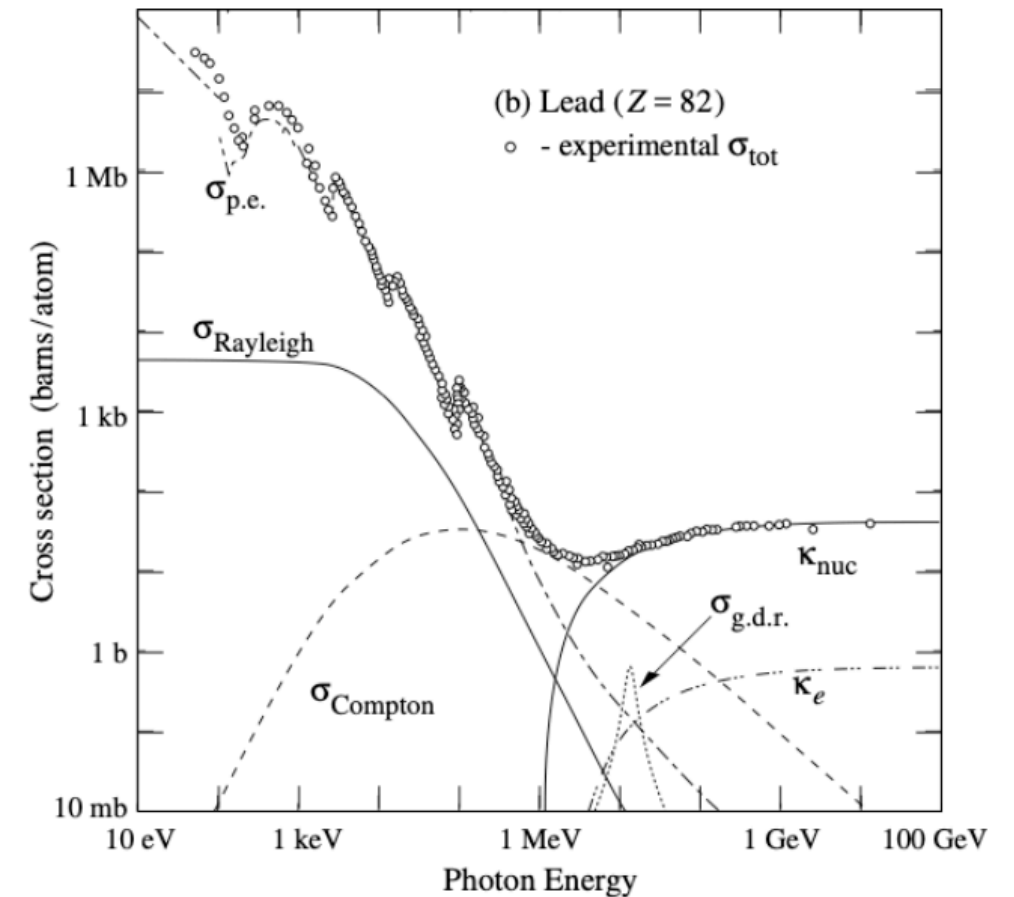
EM Overview

♦ Gamma processes

- ✧ γ conversion into e^+e^- pair
- ✧ Compton scattering
- ✧ Photoelectric effect
- ✧ Rayleigh scattering
- ✧ Gamma-nuclear interaction in hadronic library

♦ Charged particles processes

- ✧ Ionization
- ✧ Coulomb scattering
- ✧ Bremsstrahlung
- ✧ Production of e^+e^- pair
- ✧ Positron annihilation
- ✧ Nuclear interaction in hadronic library



◆ Standard

- ✧ γ , e up to 100 TeV
- ✧ hadrons up to 100 TeV
- ✧ ions up to 100 TeV

◆ Muons

- ✧ up to 1 PeV
- ✧ energy loss propagator

◆ X-rays

- ✧ Cherenkov, transition, synchrotron

◆ High-energy

- ✧ processes at high energy ($E > 10\text{GeV}$), for example γ to $\mu^+\mu^-$ pairs, e^-e^+ to π^- and π^+
- ✧ physics for exotic particles

◆ Polarisation

- ✧ models/processes for polarized beam

◆ Low-energy

- ✧ Livermore library γ , e^- from 10 eV up to 1 GeV
- ✧ Livermore library based polarized processes
- ✧ PENELOPE 2008 code rewrite γ , e^- , e^+ from 250 eV up to 6 GeV
- ✧ hadrons and ions up to 1 GeV
- ✧ atomic de-excitation (fluorescence + Auger)

◆ DNA

- ✧ Geant4 DNA modes and processes
- ✧ Micro-dosimetry models for radiobiology
- ✧ from 0.025 eV to 10 MeV
- ✧ many of them material specific (water)

◆ Adjoint

- ✧ sub-library for reverse Monte Carlo simulation from the detector of
- ✧ interest back to source of radiation

◆ Utils

- ✧ general EM interfaces and helper classes

Software Design of the EM packages

- ◆ **Uniform, coherent design approach over the different EM sub-parts**
 - ✦ Standard and low-energy EM models/processes can be combined

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- ✦ For example: G4ComptonScattering
- ✦ Assigned to particle types in the Physics List (G4ComptonScattering is assigned to photon)

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- ◆ **Three EM process interfaces to describe 3 set of interactions with different characteristics. All classes derived from these 3.**

- ✦ **G4VEmProcess** for discrete EM processes (e.g. Compton scattering)
- ✦ **G4VEnergyLossProcess** for the continuous-discrete ionization and bremsstrahlung photon emission
- ✦ **G4VMultipleScattering** for multiple Coulomb scattering (along a given step)

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- ◆ **A EM process can be simulated according to several models**

- ✦ Each model is described by a class that inherits from **G4VEmModel** base class
- ✦ Naming convention: **G4ModelNameProcessNameModel** (e.g. **G4KleinNishinaComptonModel** describes Compton scattering of photons described by the **Klein-Nishina** differential cross section)
- ✦ Models can be assigned to certain energy ranges and G4Regions

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- ◆ **Model classes provide the computation of:**

- ✦ Computation of interaction cross section (and stopping power if any)
- ✦ Computation/generation of the interaction final state (post-interaction kinematics, secondary production, etc.)

Registering electrons in Standard EM

```
msc: for e- SubType= 10
===== EM models for the G4Region DefaultRegionForTheWorld =====
      UrbanMsc : Emin=    0 eV  Emax=  100 TeV Nbins=240 100 eV  - 100 TeV
      StepLim=DistanceToBoundary Rfact=0.04 Gfact=2.5 Sfact=0.6 DispFlag:1 Skin=1 Llim=1 mm
```

```
eIoni: for e- XStype:3 SubType=2
      dE/dx and range tables from 10 eV  to 100 TeV in 260 bins
      Lambda tables from threshold to 100 TeV, 20 bins/decade, spline: 1
      StepFunction=(0.2, 0.1 mm), integ: 3, fluct: 1, linLossLim= 0.01
===== EM models for the G4Region DefaultRegionForTheWorld =====
      MollerBhabha : Emin=    0 eV  Emax=  100 TeV  deltaVI
```

```
eBrem: for e- XStype:4 SubType=3
      dE/dx and range tables from 10 eV  to 100 TeV in 260 bins
      Lambda tables from threshold to 100 TeV, 20 bins/decade, spline: 1
      LPM flag: 1 for E > 1 GeV, VertexHighEnergyTh(GeV)= 100000
===== EM models for the G4Region DefaultRegionForTheWorld =====
      eBremSB : Emin=    0 eV  Emax=    1 GeV  AngularGen2BS
      eBremLPM : Emin=    1 GeV Emax=  100 TeV  AngularGen2BS
```

```
ePairProd: for e- XStype:1 SubType=4
      dE/dx and range tables from 10 eV  to 100 TeV in 260 bins
      Lambda tables from threshold to 100 TeV, 20 bins/decade, spline: 0
      Sampling table 25x1001; from 0.1 GeV to 100 TeV
===== EM models for the G4Region DefaultRegionForTheWorld =====
      ePairProd : Emin=    0 eV  Emax=  100 TeV  ModifiedMephi
```

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            Sampling table 25x1001; from 0.1 GeV to 100 TeV
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```

Gamma Processes

◆ Main processes (low to high energy)

- ❖ Photoelectric effect
- ❖ Rayleigh scattering should not be neglected if an accurate dosimetry simulation is needed
- ❖ Compton scattering
- ❖ Gamma absorption by Nuclei (giant dipole resonance)
- ❖ γ conversion into e^+e^- pair

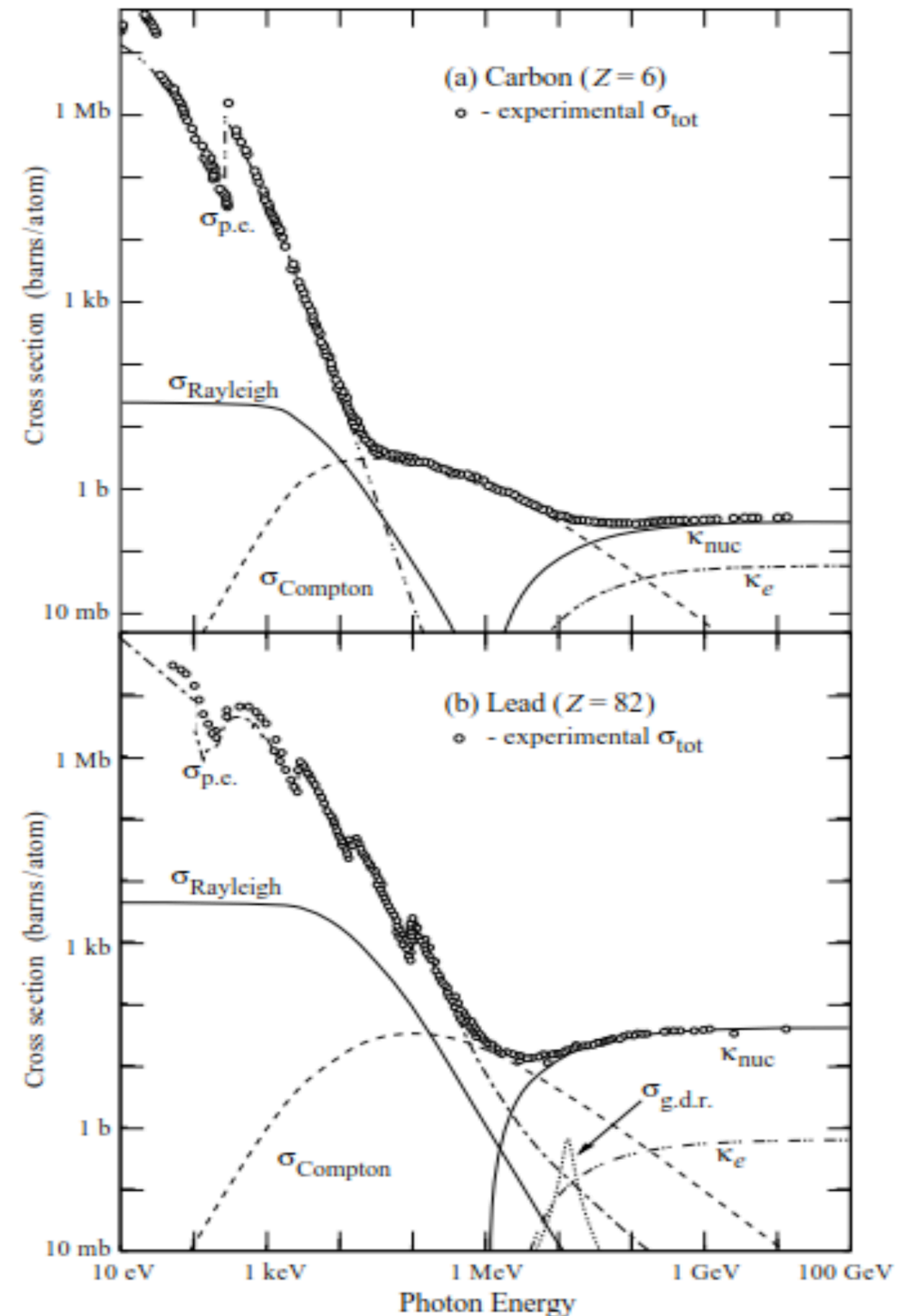
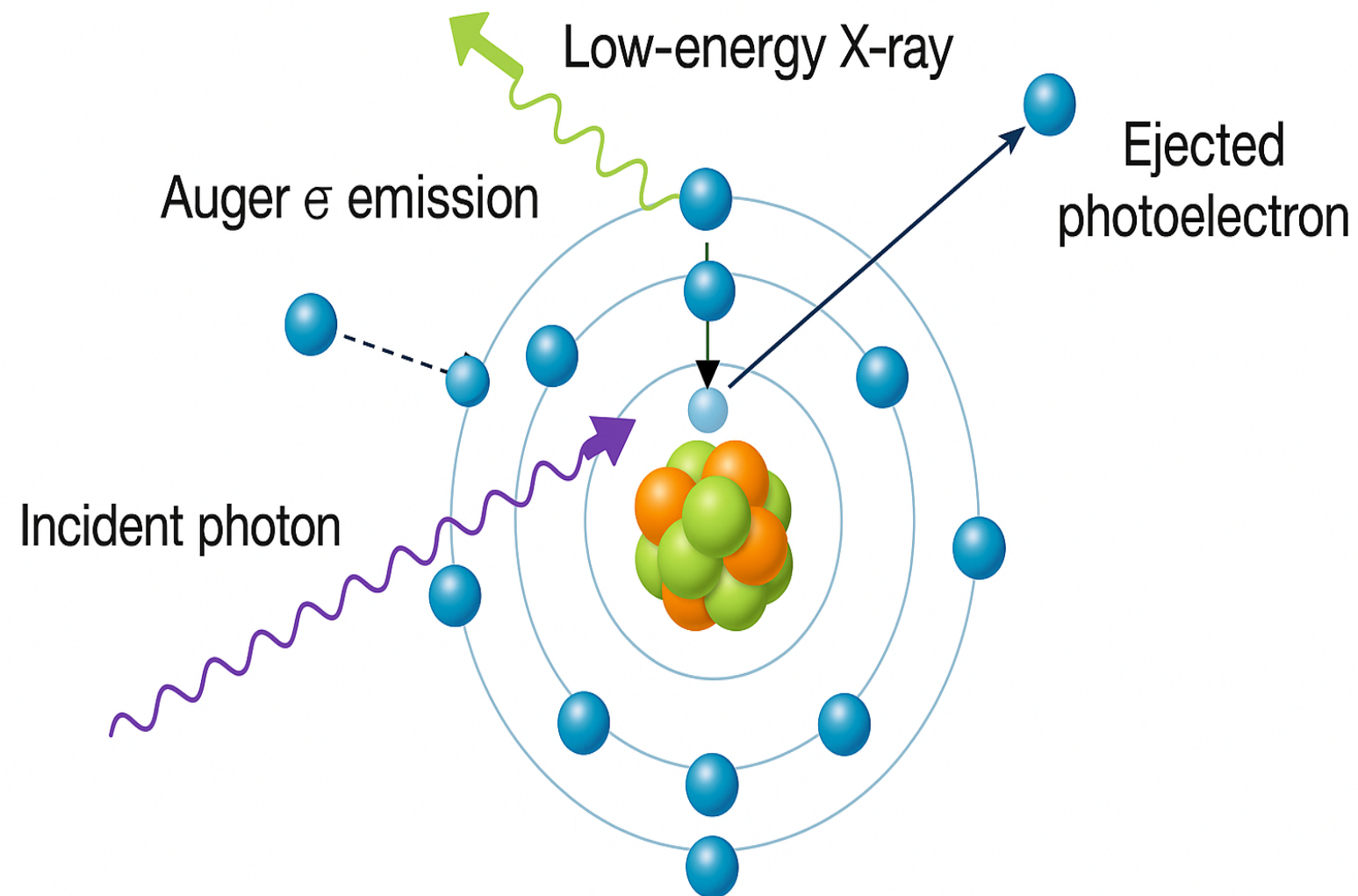


Photo-electric absorption and atomic de-excitation

- ◆ Photo-electric absorption leaves the atom in an excited state
- ◆ Vacancy in the ionized shell
- ◆ Decay through a cascade of radiated and non radiated transitions
- ◆ Emission of X-rays and other electrons



Atomic de-excitation

- ♦ Atomic de-excitation is **initiated** by other EM physics interactions
 - ✦ Photoelectric effect, ionization (by e- or ions), Compton scattering, ...
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- ♦ The **EADL** (Evaluated Atomic Data Library) contains transition probabilities: Decay through a cascade of radiated and non radiated transitions
 - ✧ Radiative transition characteristic X-ray emission (fluorescence photon emission)
 - ✧ Auger e- emission: initial and final vacancies are in different shells
 - ✧ Coster-Kronig e- emission: initial and final vacancies are in the same shells

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 - ✧ Coster-Kronig e- emission: initial and final vacancies are in the same shells
- ♦ De-excitation has common interface so is compatible with both the standard and the low-energy EM physics categories
 - ✧ It can be enabled and controlled by UI command (before initialization)

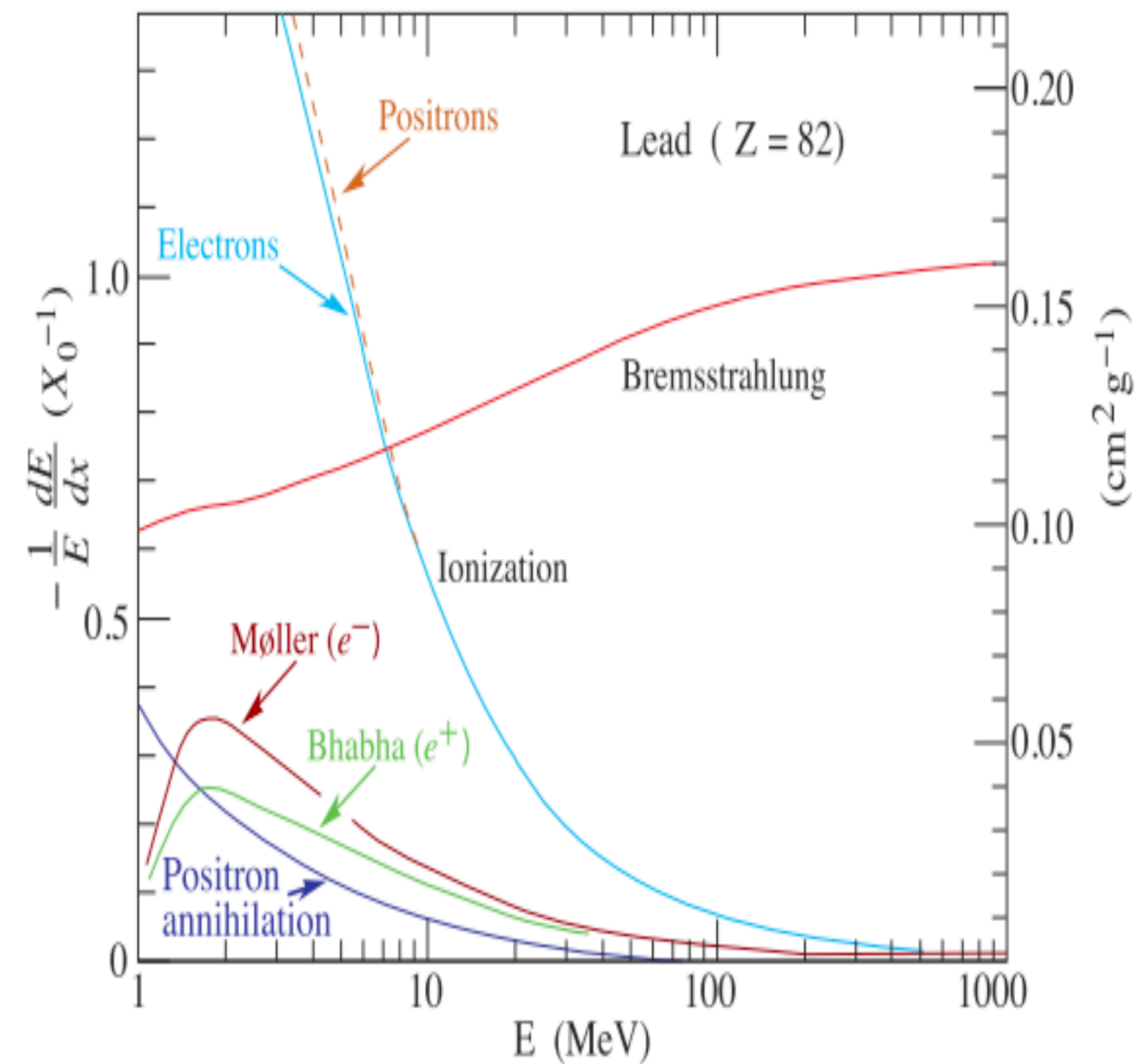
```
/process/em/fluor true  
/process/em/auger true  
/process/em/pixe true  
/run/initialize
```

- ✧ The fluorescence transition is active by default in some EM physics constructor while others (Auger, PIXE) are not

Charged particles processes

◆ Charged particles processes (low to high energy)

- ✧ Positron annihilation
- ✧ Ionization
- ✧ Bremsstrahlung
- ✧ Coulomb scattering
- ✧ Production of e^+e^- pair



Difference between electrons and positrons is significant for low energy but practically negligible above critical energy

Charged particles: simulation of g4step

- ◆ Values of mean dE/dx , range, cross section of δ -electron production, and bremsstrahlung are pre-computed at initialization stage of Geant4 and are stored in a `G4PhysicsTable`

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- ◆ At run time, for each simulation **g4step**
 1. a spline interpolation of tables is used to get the mean energy loss
 2. a call to the the general interface to a fluctuation model is `G4VEmFluctuationModel`
 3. a sampling of the energy loss fluctuation is performed

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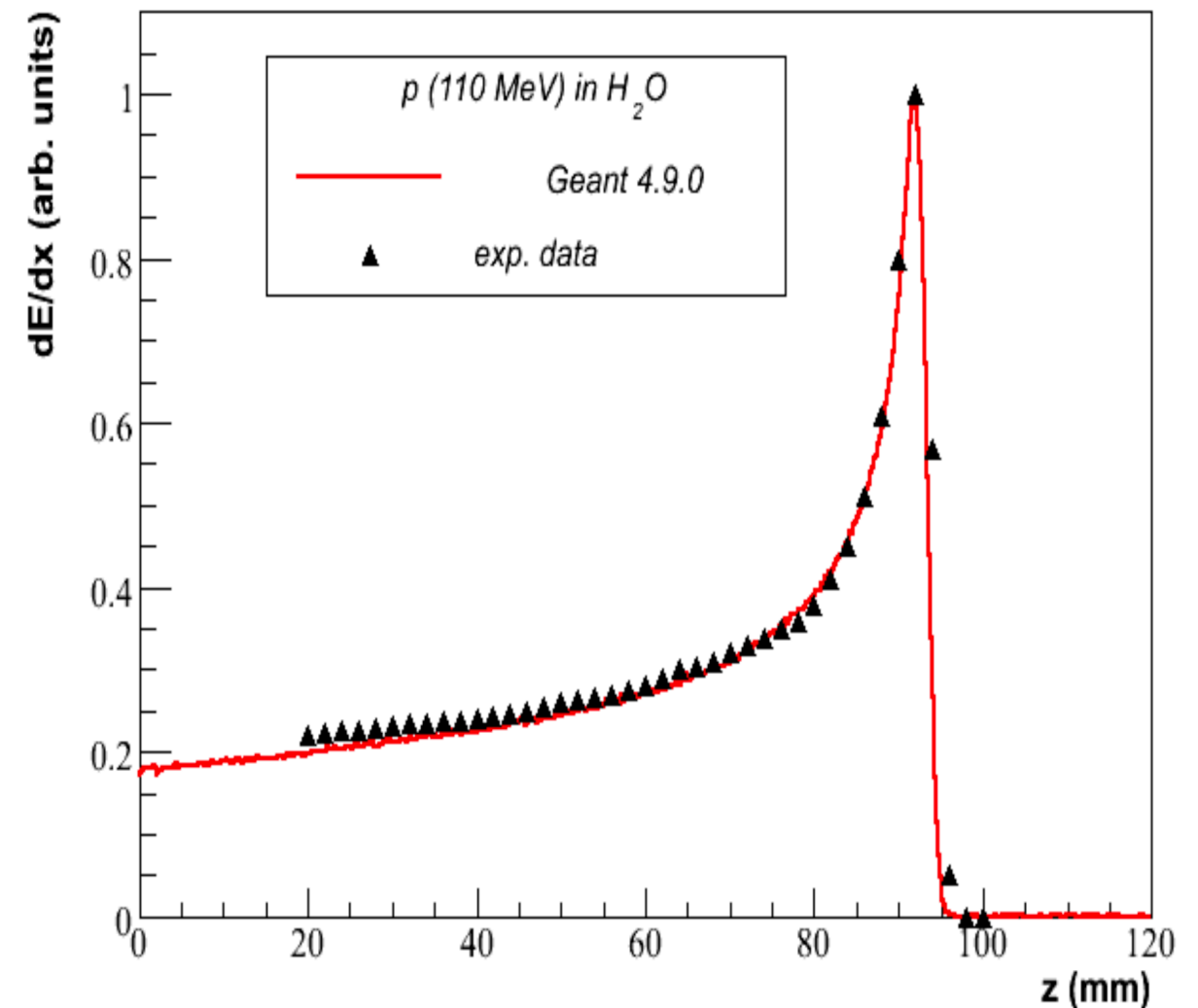
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 3. a sampling of the energy loss fluctuation is performed
- ◆ The cross sections of δ -electron production and bremsstrahlung are used to sample production above the threshold T_{CUT} at `PostStep`
- ◆ If atomic de-excitation is active, then fluorescence and Auger electron production is sampled `AlongStep` and `PostStep`

1: Hadrons and ion ionization: get mean energy loss

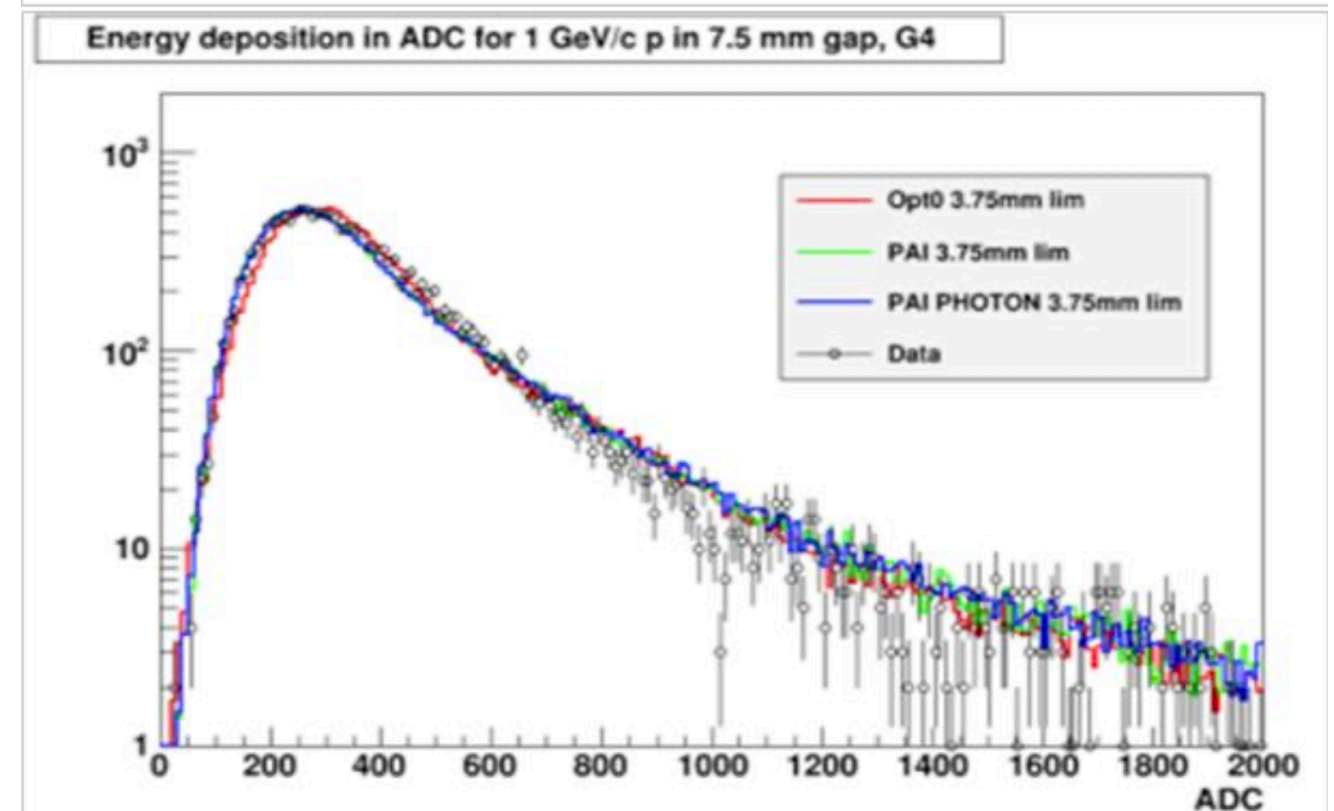
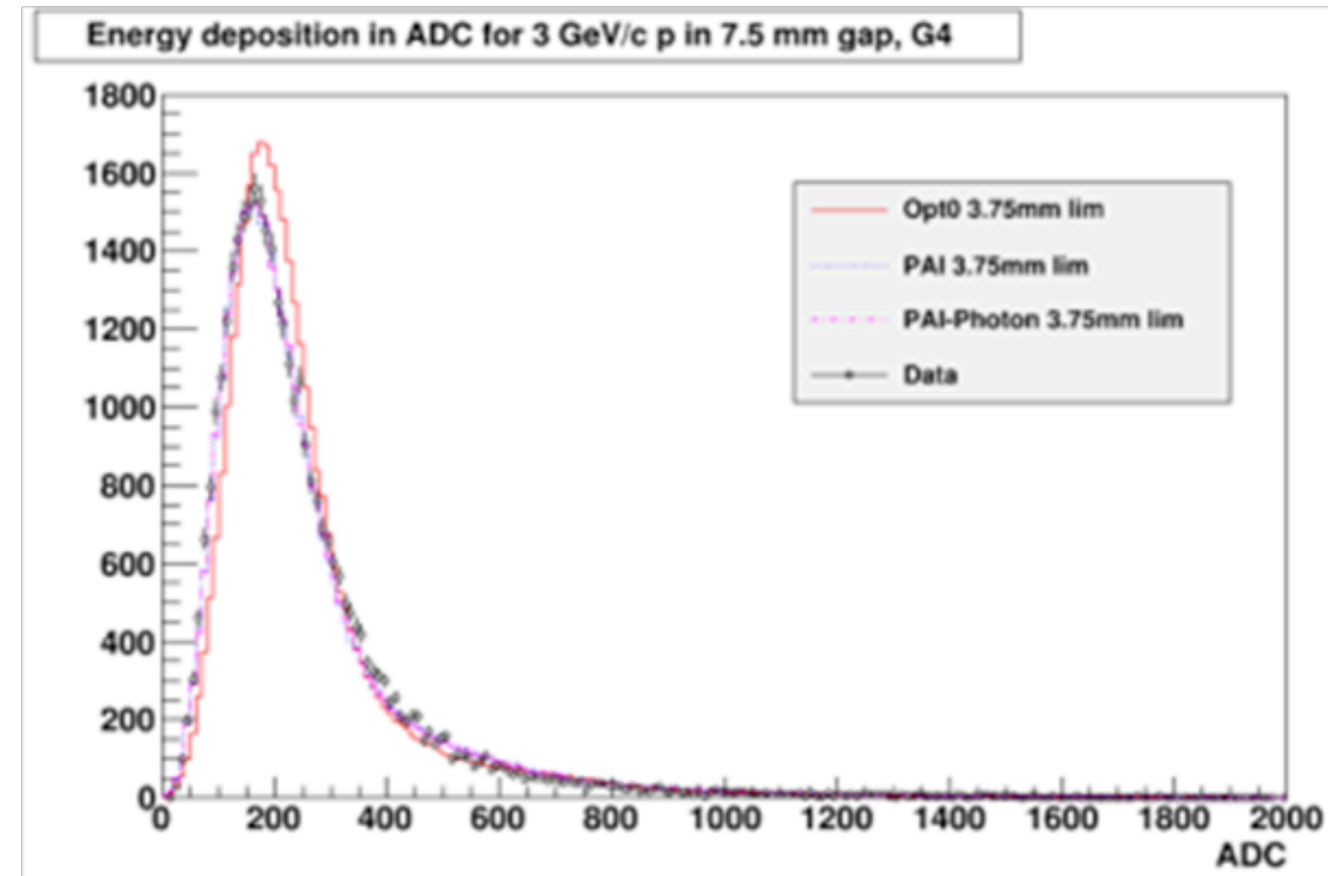
$$-\frac{dE}{dx} = 4\pi N_e r_0^2 \frac{z^2}{\beta^2} \left(\ln \frac{2m_e c^2 \beta^2 \gamma^2}{I} - \frac{\beta^2}{2} \left(1 - \frac{T_c}{T_{\max}} \right) - \frac{C}{Z} + \frac{G - \delta - F}{2} + zL_1 + z^2 L_2 \right)$$

- ◆ C: shell correction
- ◆ G: Mott correction
- ◆ δ density correction
- ◆ F: finite size correction
- ◆ L_1 : Barkas correction
- ◆ L_2 : Block correction
- ◆ Nuclear stopping
- ◆ Ion effective range
- ◆ Scaling for heavy particles / ions



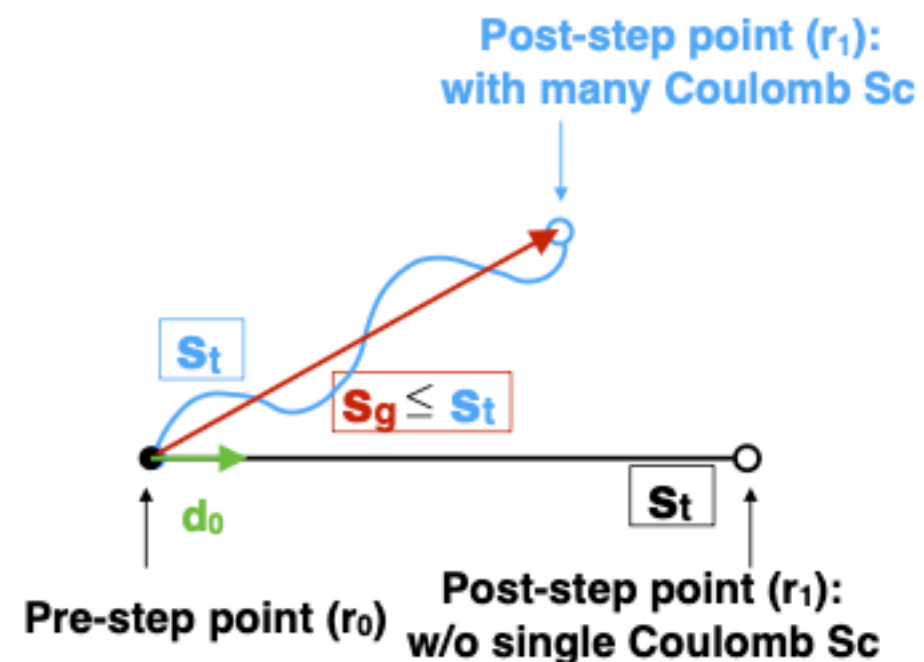
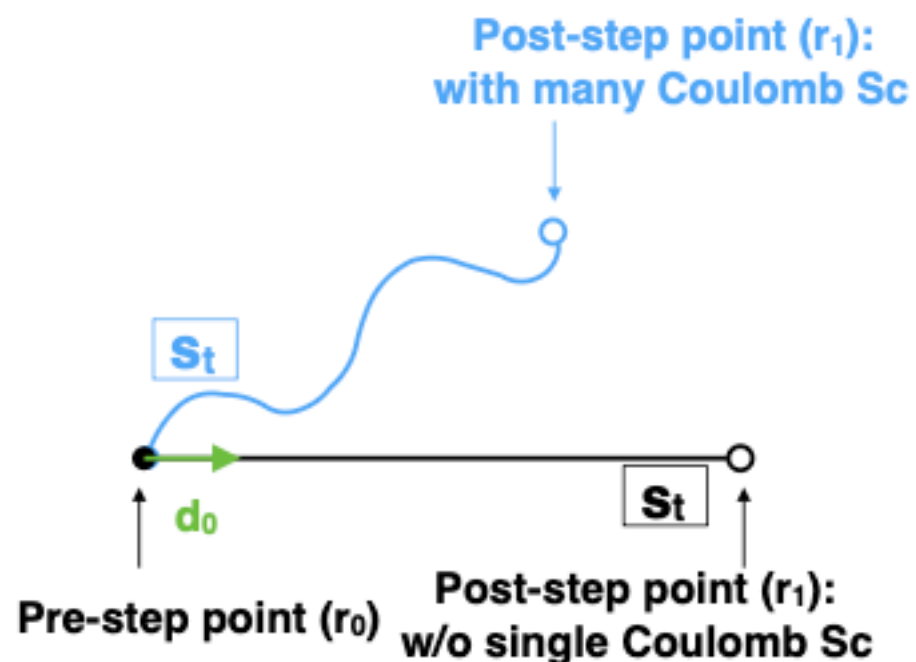
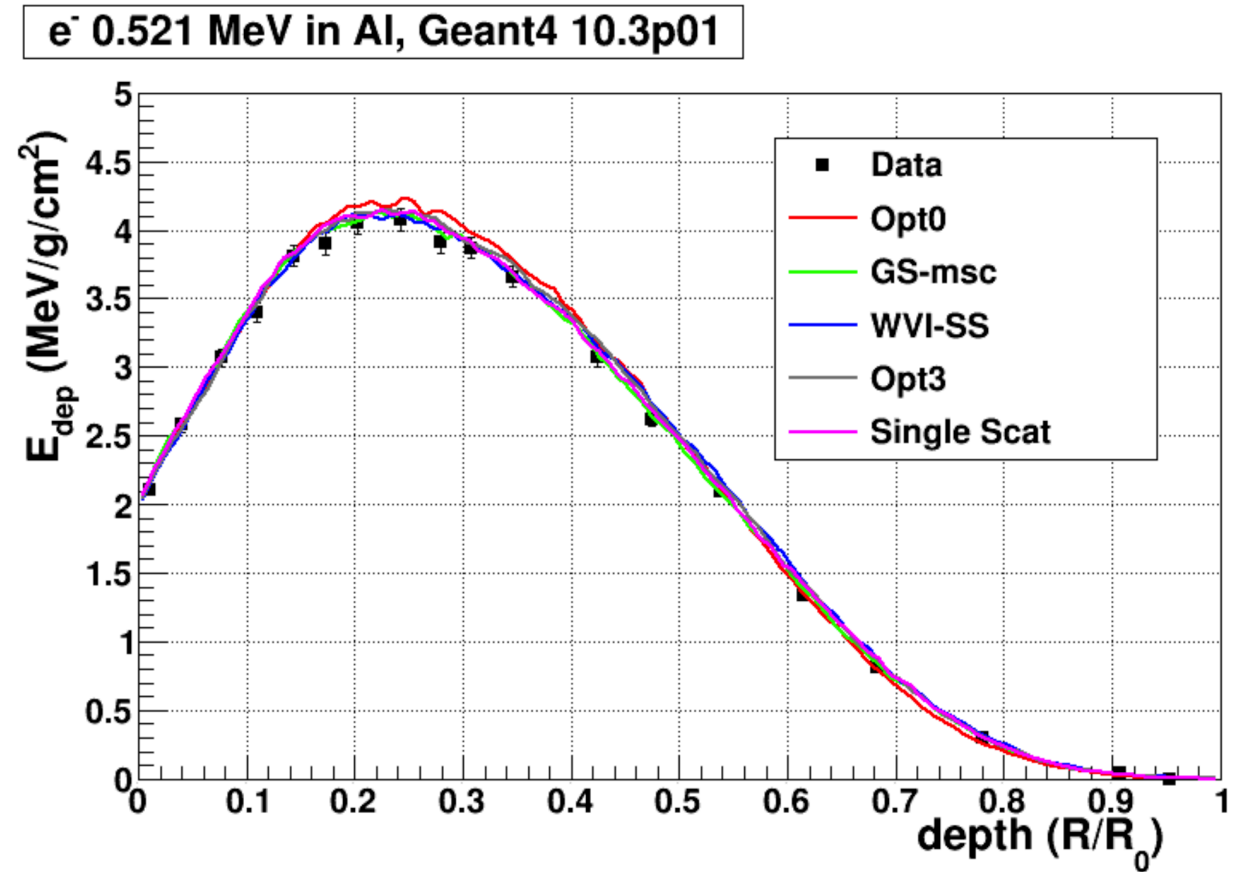
2, 3: Fluctuations models

- ♦ **Urban model** based on a simple model of particle-atom interaction
 - ❖ Atoms are assumed to have only two energy levels: E_1 and E_2
 - ❖ Particle-atom interaction can be:
 - ❖ an excitation of the atom with energy loss $E = E_1 - E_2$
 - ❖ an ionization with energy loss distribution $g(E) \approx 1/E^2$
- ♦ **PAI model** uses photo absorption cross section data
 - ❖ Relativistic model
 - ❖ Energy transfers are sampled with production of secondary e^- or γ
 - ❖ Very slow model
 - ❖ Takes into account each ionization collision



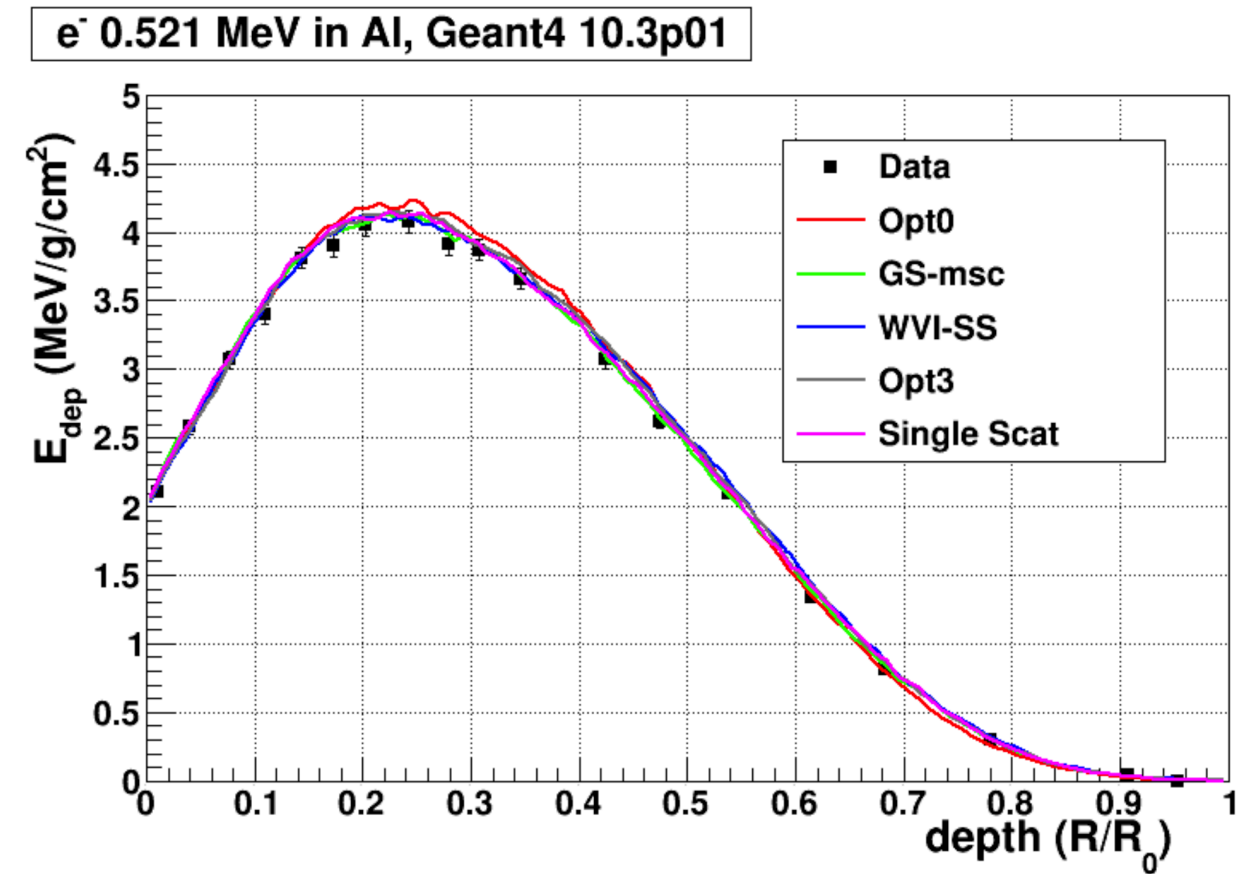
Electron/positron Multiple Scattering Models

- ◆ **Coulomb scattering:** elastic scattering of charged particles on the atomic potential
- ◆ **Event-by-event** modeling of elastic scattering is feasible only if the mean number of interactions per track is below few hundred
- ◆ This limits the applicability of the detailed simulation model only for electrons with relatively low kinetic energies (up to $E_{\text{kin}} \sim 100$ keV) or thin targets
- ◆ **MSC model:** give the final position and direction of the particle while try to minimize the possible errors due to geometrical constraints



Electron/positron Multiple Scattering Models

- ◆ **MSC model:** give the final position and direction of the particle while try to minimize the possible errors due to geometrical constraints
- ◆ Can be controlled with UI command (in increasing accuracy and simulation time)



```
/process/msc/StepLimit UseSafety  
/process/msc/StepLimit UseDistanceToBoundary  
/process/msc/StepLimit UseSafetyPlus
```

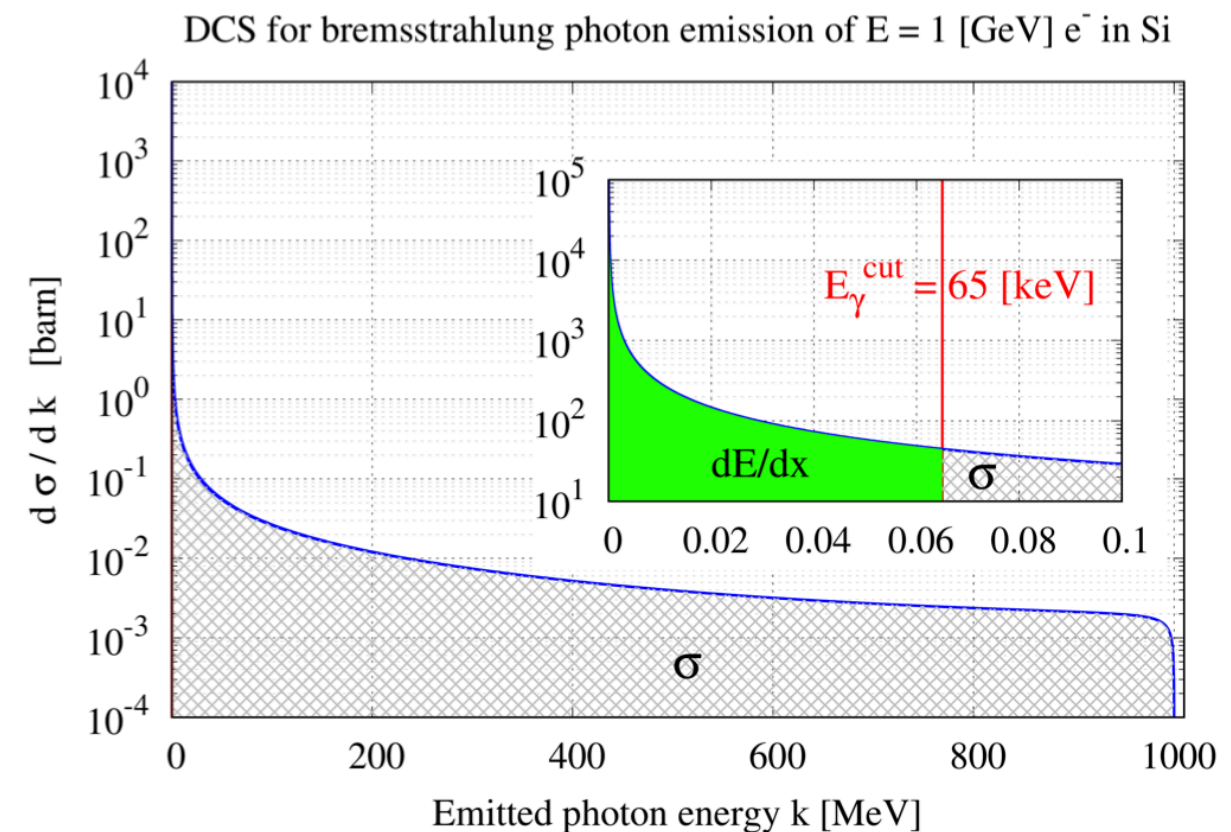
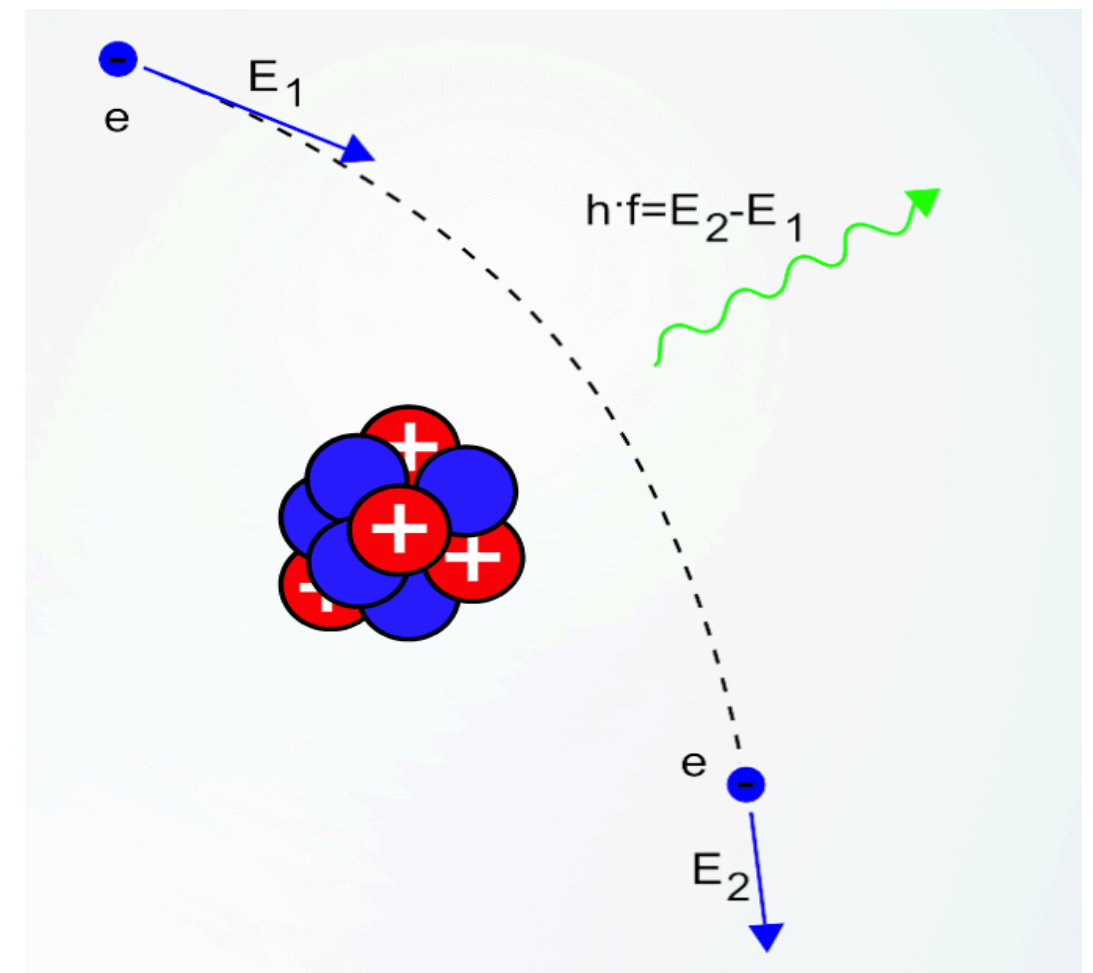
these are for e^-/e^+ but different UI command available for muons and hadrons

Production Threshold

For Bremsstrahlung

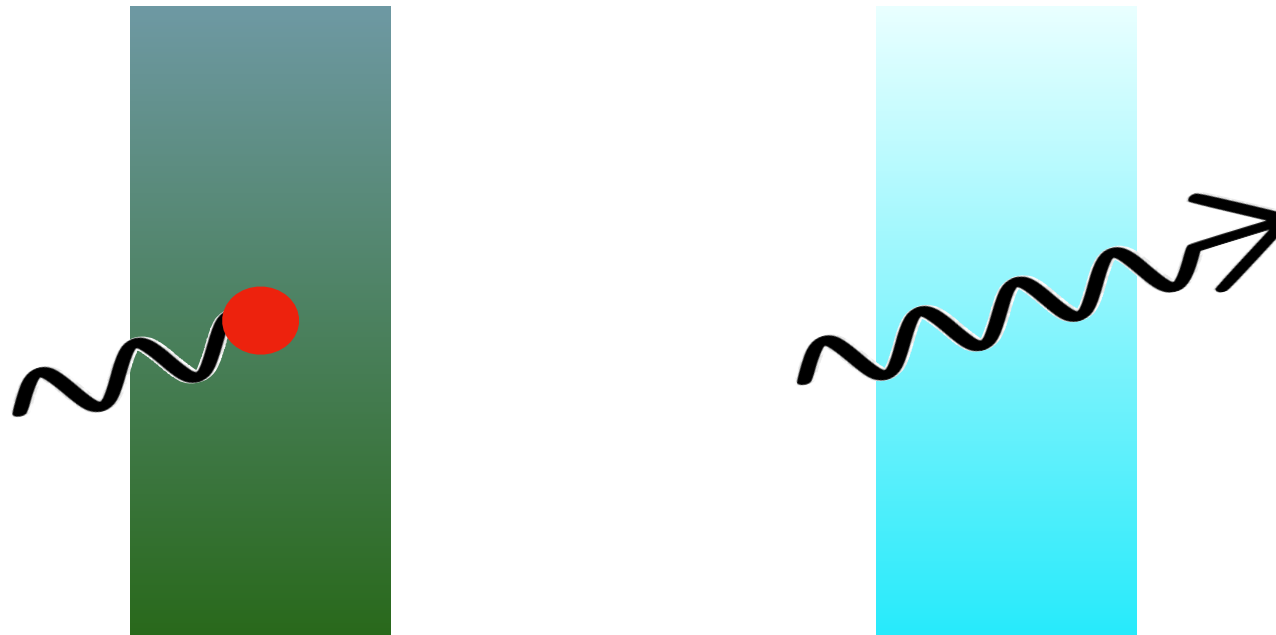
- ♦ Low energy photons (k) are emitted with high rate:
DCS $\sim 1/k$
- ♦ The generation and tracking of all these low energy photons would not be feasible (CPU time)
- ♦ Low energy photons has a very small absorption length
- ♦ **If the detector spacial resolution is worse than this length** then the following scenarios are exactly the same in terms of final result of a simulation:
 - Generating and tracking these low energy photons till all their energy will be deposited
 - Depositing their corresponding energy at the creation point (at a trajectory point)

**B is equivalent to A but much faster.
We can control these situations with a
production threshold cut.**



Energy vs Length

- ✧ It may seem more intuitive at first for users to provide a threshold in ENERGY
- ✧ But the same energy will translate to different lengths (absorption length, range) in different materials: a 10 keV gamma has **very different absorption length in Pb or in Ar gas**



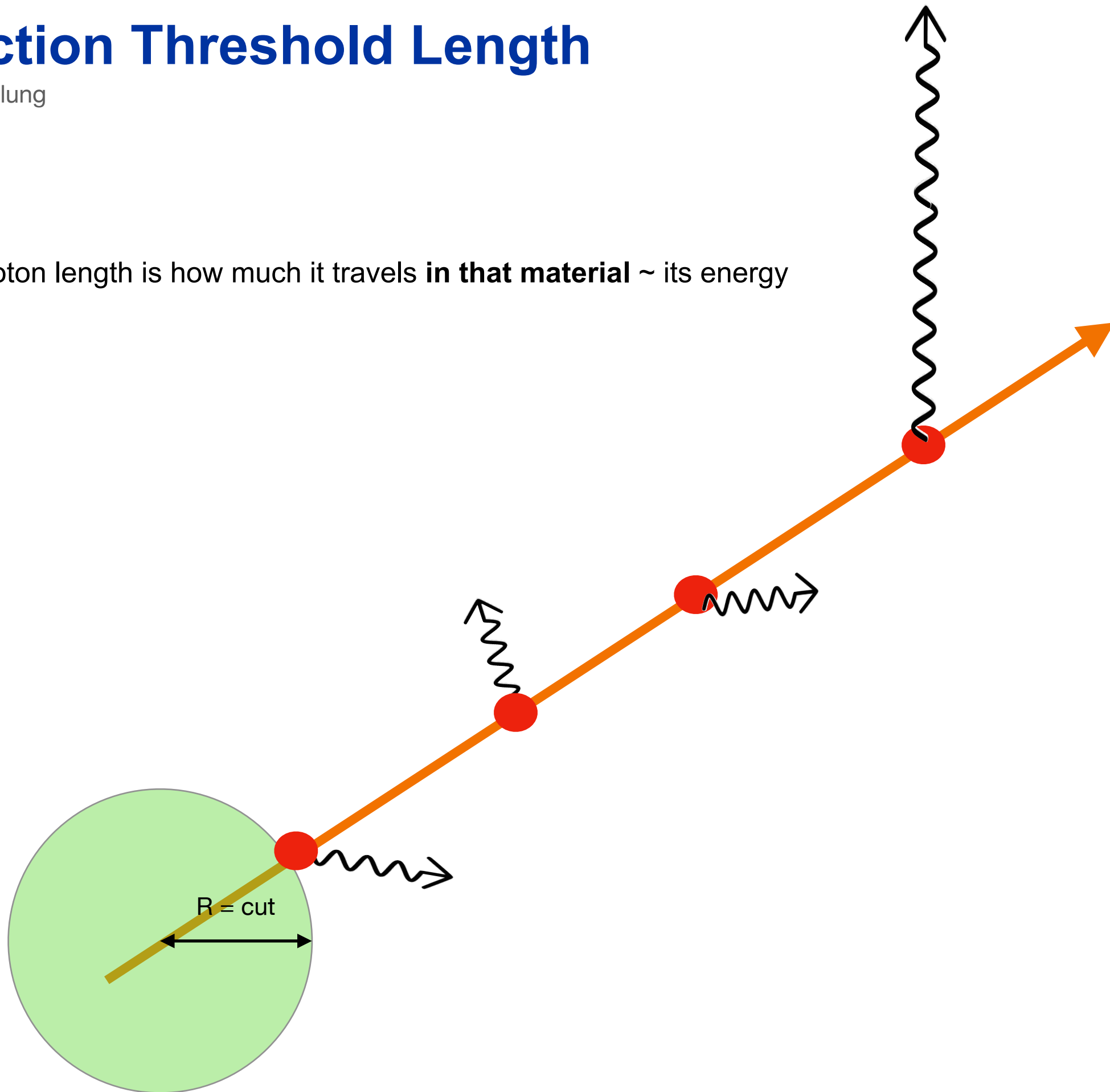
- ✧ Moreover, the same energy will translate to different lengths depending on the particle type (gamma => absorption length; e-/e+ => range) even in the same material: range of a 10 keV e- in Si is few micron while the absorption length of a 10 keV gamma in Si is few cm
- ✧ If we wanted to use energy threshold, we would need values per material and per particle!

A cut in length is more intuitive (after you think about it a bit) and directly related to detector dimensions

Production Threshold Length

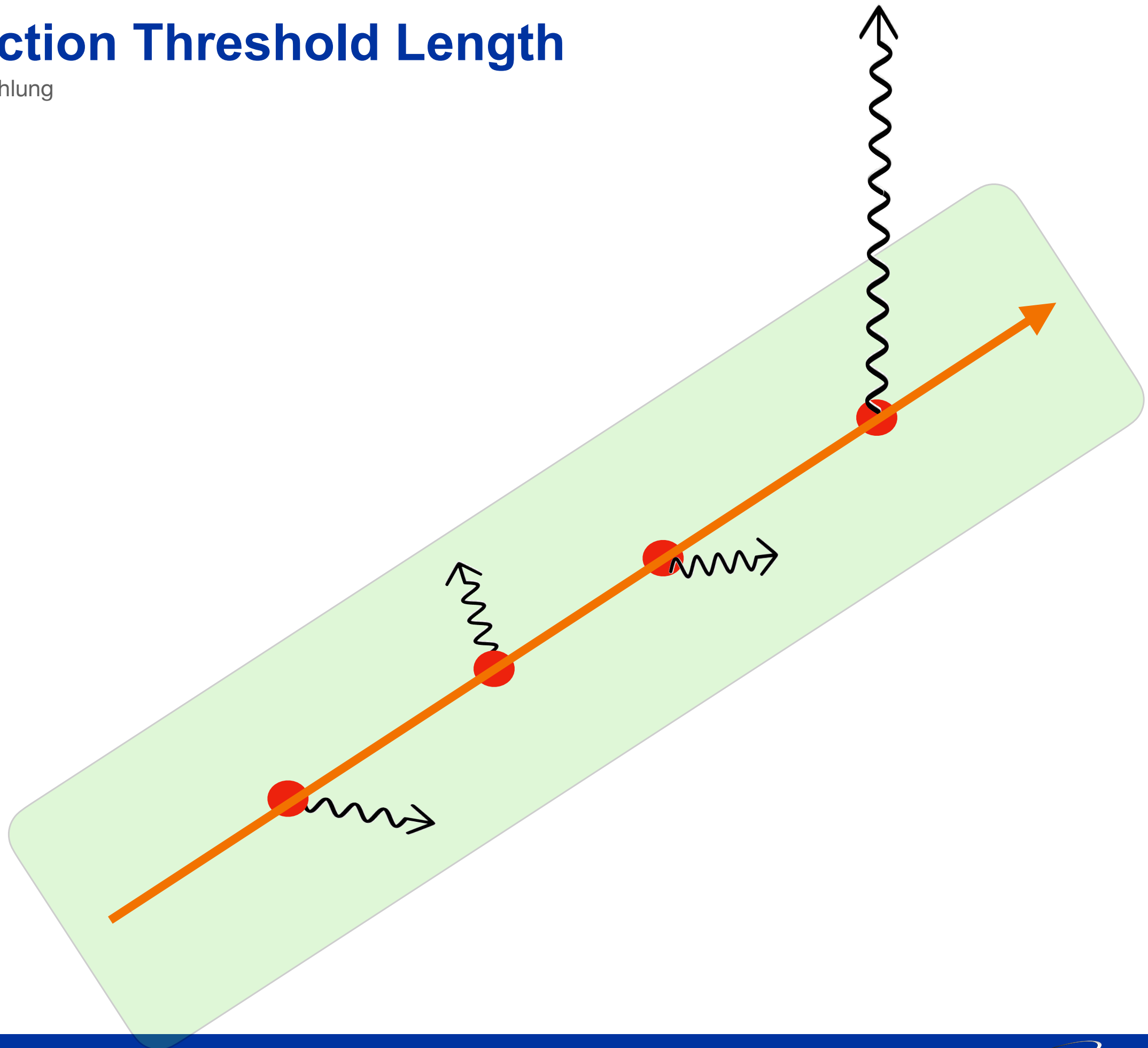
For Bremsstrahlung

Here the photon length is how much it travels **in that material** ~ its energy



Production Threshold Length

For Bremsstrahlung



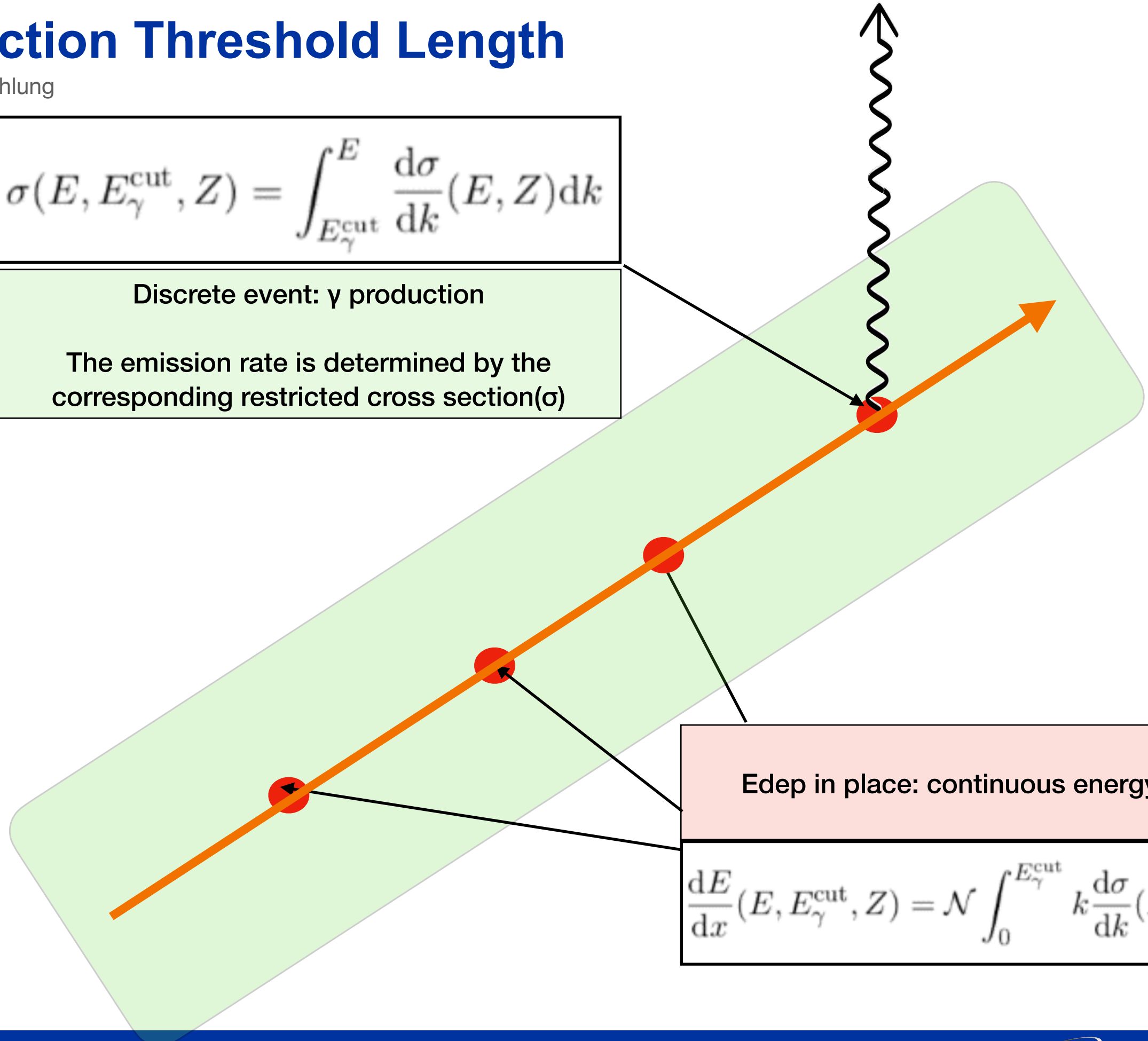
Production Threshold Length

For Bremsstrahlung

$$\sigma(E, E_{\gamma}^{\text{cut}}, Z) = \int_{E_{\gamma}^{\text{cut}}}^E \frac{d\sigma}{dk}(E, Z) dk$$

Discrete event: γ production

The emission rate is determined by the corresponding restricted cross section(σ)



Production Threshold

- ♦ The corresponding **Edep** energy (that would have been taken away from the primary) is accounted as **CONTINUOUS energy loss** of the primary particle along its trajectory
- ♦ **Same applies to ionization with the difference:**
 - ❖ secondary gamma → secondary e- production threshold
 - ❖ absorption length → range
- ♦ Range and absorption production length is **internally translated to energies** at initialization, taking care of material and particle dependency
- ♦ Production threshold defined for gamma, e-, e+ and proton secondary particle:
 - ❖ gamma production threshold is used in bremsstrahlung while the e- in ionization
 - ❖ e+ production threshold might be used in case of e-/e+ pair production
 - ❖ proton production threshold is used as a kinetic energy threshold for nuclear recoil in case of elastic scattering of all hadrons and ions
- ♦ Production threshold are not mandatory, but many physics simulation would not be feasible w/o them

- ◆ User needs to provide them in length (with a default value of 0.7 mm for the reference physics lists)
- ◆ UI commands to define cuts:

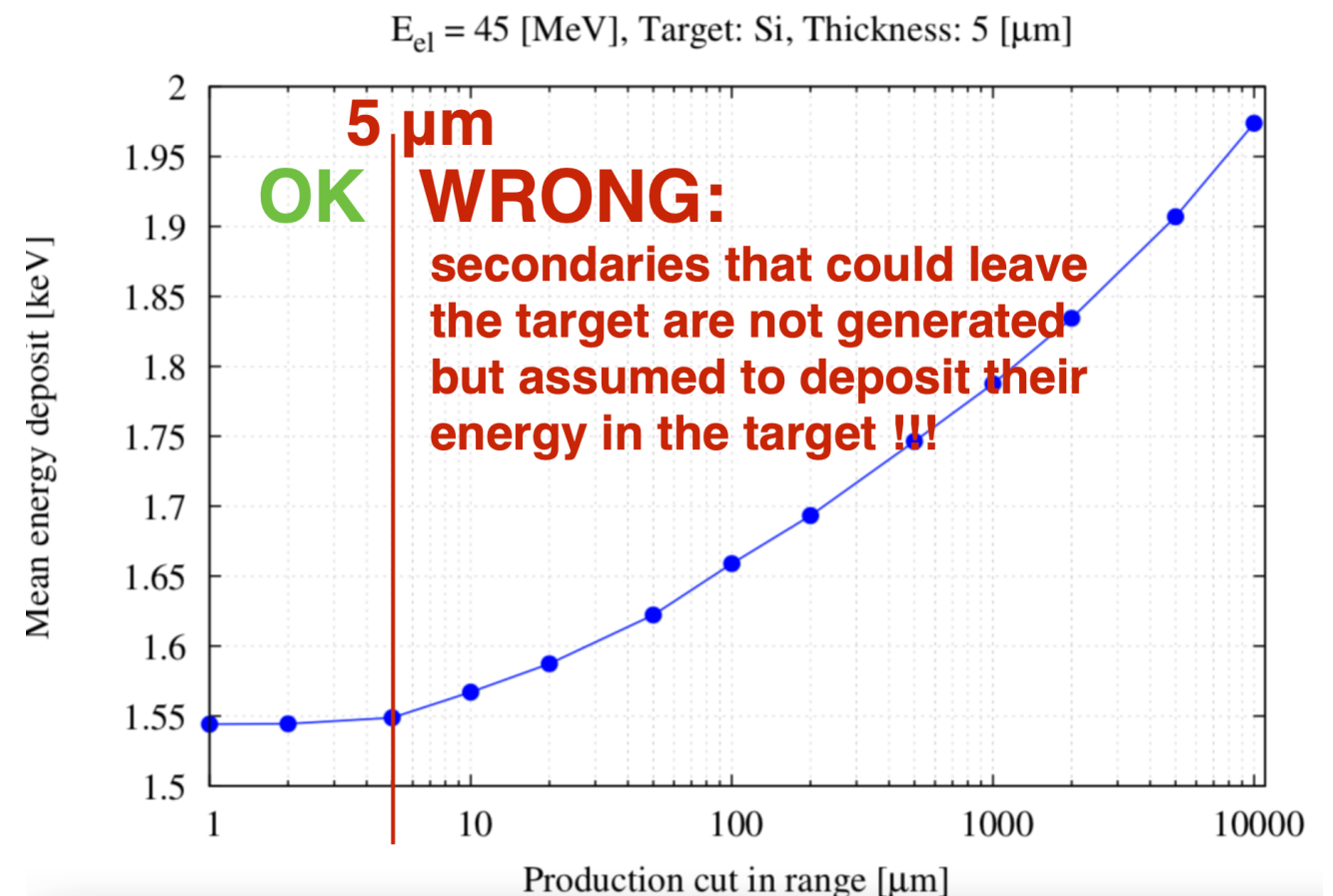
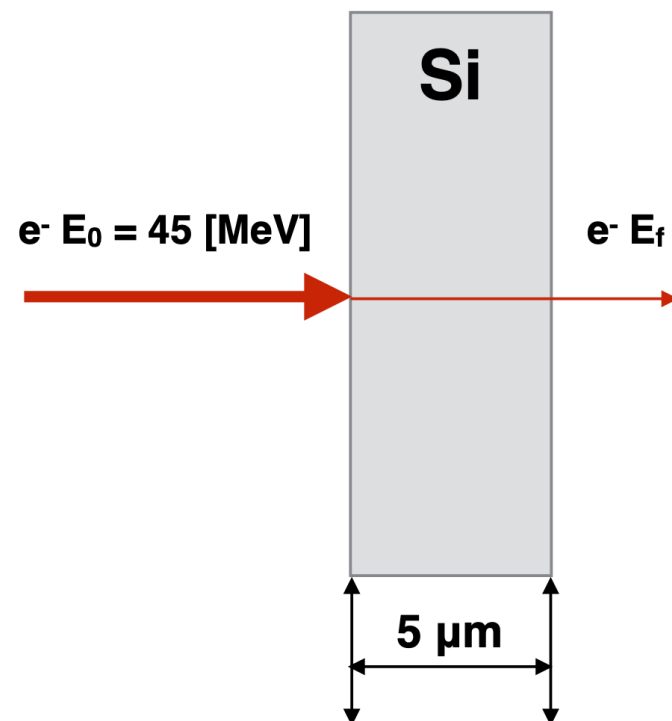
```
/run/setCut 0.1 mm  
/run/setCutForAGivenParticle e- 0.01 mm  
/run/setCutForRegion GasDetector 0.001 mm  
/cuts/setLowEdge 500 eV
```

Production Threshold

What you need to do

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- ◆ UI commands to define cuts:

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Production Threshold per detector region

- ✦ Different parts of a complex detector might require modeling with different level of details and have different spacial resolution
- ✦ Different detector **G4Regions** can be defined and a set of **G4LogicalVolumes** can be associated to such regions
- ✦ Different secondary production threshold values (as well as G4UserLimits) can be assigned to different detector region
- ✦ In the `DetectorConstruction::Construct()` method (e.g. examples/ extended/ electromagnetic/TestEm9)

```
// Production threshold for "Our-Region"  
auto* cuts = new G4ProductionCuts;  
cuts->SetProductionCut(0.5*mm); // same cuts for gamma, proton, e- and e+  
cuts->SetProductionCut(0.2*mm, G4ProductionCuts::GetIndex("e-"));
```

```
// Create region and assign cuts  
auto* region = new G4Region("Our-Region");  
region->SetProductionCuts(cuts);
```

```
// Assign volume to region: all volumes will have these cuts  
region->AddRootLogicalVolume(my_specific_logical_volume);
```


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```
Index : 0      used in the geometry : Yes
Material : Galactic
Range cuts      : gamma 700 um      e- 700 um      e+ 700 um      proton 700 um
Energy thresholds : gamma 990 eV      e- 990 eV      e+ 990 eV      proton 70 keV
Region(s) which use this couple :
    DefaultRegionForTheWorld

Index : 1      used in the geometry : Yes
Material : G4_Pb
Range cuts      : gamma 700 um      e- 700 um      e+ 700 um      proton 700 um
Energy thresholds : gamma 94.5861 keV      e- 1.00386 MeV      e+ 951.321 keV      proton 70 keV
Region(s) which use this couple :
    DefaultRegionForTheWorld

Index : 2      used in the geometry : Yes
Material : G4_lAr
Range cuts      : gamma 100 um      e- 100 um      e+ 100 um      proton 100 um
Energy thresholds : gamma 2.00482 keV      e- 82.9692 keV      e+ 81.8616 keV      proton 10 keV
Region(s) which use this couple :
    Our-Region
```

Standard EM Physics Constructors

List of particles for which EM physics processes are defined

- ◆ Photon, leptons, meson and baryons:

γ , $e^\pm$, $\mu^\pm$, $\pi^\pm$, $K^\pm$, p , $\Sigma^\pm$, Ξ^- , Ω^- , $\text{anti}(\Sigma^\pm, \Xi^-, \Omega^-)$

- ◆ Heavy leptons & heavy mesons, charmed baryons:

$\tau^\pm$, $B^\pm$, $D^\pm$, $Ds^\pm$, Λ_c^+ , Σ_c^+ , Σ_c^{++} , Ξ_c^{++} , $\text{anti}(\Lambda_c^+, \Sigma_c^+, \Sigma_c^{++}, \Xi_c^+)$

- ◆ Light nuclei:

d , t , ^3He , ^4He , generic-ion, $\text{anti}(d, t, ^3\text{He}, ^4\text{He})$

- ◆ 12 light hyper- and anti-hyper- nuclei

Each particle has its own **G4ProcessManager** that maintains the list of assigned processes (e.g., ionization, multiple scattering, decay)

Standard EM Physics Constructors

- ◆ The modular Physics Lists (G4VModularPhysicsList) allows to build up a complete physics list from “physics modules”
- ◆ A given “physics module” handles a well defined category of physics (e.g. EM physics, decay physics, etc.) as a sub-set of a complete physics list
- ◆ **G4VPhysicsConstructor** is the Geant4 interface to describe such sub-sets of physics
- ◆ Several **EM G4VPhysicsConstructor**, i.e. pre-defined EM sub-set of a complete physics list, are available in Geant4
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```
auto particleIterator = GetParticleIterator();
particleIterator->reset();
while( (*particleIterator)() ){

    G4ParticleDefinition* particle = particleIterator->value();

    auto* pmanager = particle->GetProcessManager();

    // Attach processes here
    pmanager->AddProcess(new G4eIonisation, ordInActive, ordAlongStep, ordPostStep);

}
```

Standard EM Physics Constructors for HEP

- ◆ SAME: Description of Coulomb scattering
 - ✧ $e^\pm$: Urban - MSC model below 100 [MeV] and the Wentzel - WVI + Single scattering (mixed simulation) model above 100 [MeV]
 - ✧ muon and hadrons: Wentzel - WVI + Single scattering (mixed simulation) model
 - ✧ ions: Urban - MSC model
- ◆ Different MSC stepping algorithms and/or parameters: speed v.s. accuracy

Constructor	Components	Comment
EmStandardPhysics	Defaults (FTFP_BERT)	for ATLAS and other HEP simulation applications
G4EmStandardPhysics_option1	Fast: due to simpler MSC step limitation, cuts used by photon processes (FTFP_BERT_ EMV)	similar to one used by CMS; good for crystals but not good for sampling calorimeters (i.e. with more detailed geometry)
G4EmStandardPhysics_option2	Experimental: similar to option1 with updated photoelectric model but no-displacement in MSC (FTFP_BERT_ EMX)	similar to one used by LHCb

Standard EM Physics Constructors: more precise

- ♦ The primary goal is more the physics accuracy over the speed
- ♦ Combination of standard and low-energy EM models for more accurate physics description
- ♦ More accurate models for $e^\pm$ MSC (Goudsmit-Saunderson(GS)) and/or more accurate stepping algorithms (compared to HEP)
- ♦ Stronger continuous step limitation due to ionization (as others given per particle groups)
- ♦ Recommended for more accuracy sensitive applications: medical (hadron/ion therapy), space

Constructor	Components	Comment
G4EmStandardPhysics_option3	Urban MSC model for all charged particles (FTFP_BERT_EMY)	proton/ion therapy
G4EmStandardPhysics_option4	most accurate combination of models (particle type and energy) GS MSC (FTFP_BERT_EMZ)	The most accurate EM physics
G4EmLivermorePhysics	Livermore gamma and e- ionisation models GS MSC (FTFP_BERT_LIV)	Recommended for cross-checks of option4
G4EmPenelopePhysics	Penelope gamma and $e^\pm$ ionisation and bremsstrahlung models GS MSC (FTFP_BERT_PEN)	Recommended for cross-checks of option4

Extra experimental constructors are available in Geant4 examples

EM parameters of any EM physics list may be accessed or modified at initialization of Geant4 using:

◆ UI commands

```
/process/eLoss/verbose 0  
/process/eLoss/maxKinEnergyCSDA 100 TeV
```

```
/process/em/deexcitationIgnoreCut true  
/process/eLoss/UseAngularGenerator true  
/process/em/lowestElectronEnergy 50 eV  
/process/em/lowestMuHadEnergy 100 keV
```

```
/process/em/printParameters
```


EM parameters of any EM physics list may be accessed or modified at initialization of Geant4 using:

- ◆ C++ interface to the **G4EmParameter** (EM options) class

`PhysListEmStandard.cc` in `extended/electromagnetic/TestEm0`

```
PhysListEmStandard::PhysListEmStandard(const G4String& name) :  
    G4VPhysicsConstructor(name) {  
  
    G4EmParameters* param = G4EmParameters::Instance();  
    param->SetDefaults();  
    param->SetVerbose(0);  
    SetPhysicsType(bElectromagnetic);  
}
```

`PhysycsList.cc` in `extended/electromagnetic/TestEm0`

```
void PhysicsList::AddPhysicsList(const G4String& name) {  
  
    G4EmParameters::Instance()->SetBuildCSDARange(true);  
    G4EmParameters::Instance()->SetGeneralProcessActive(false);  
}
```

EM parameters of any EM physics list may be accessed or modified at initialization of Geant4 using:

- ◆ C++ interface to the **G4EmCalculator** (access to cross sections and stopping powers) class

`RunAction.cc` in `extended/electromagnetic/TestEm0`

```
void RunAction::CriticalEnergy()
{
    // compute e- critical energy (Rossi definition) and Moliere radius.
    // Review of Particle Physics - Eur. Phys. J. C3 (1998) page 147

    G4EmCalculator emCal;
    const G4Material* material = fDetector->GetMaterial();
    const G4double radl = material->GetRadlen();
    G4double ekin = 5 * MeV;
    G4double deioni;
    G4double err = 1., errmax = 0.001;
    G4int iter = 0, itermax = 10;
    while (err > errmax && iter < itermax) {
        iter++;
        deioni = radl * emCal.ComputeDEDX(ekin, G4Electron::Electron(), "eIoni", material);
        err = std::abs(deioni - ekin) / ekin;
        ekin = deioni;
    }
}
```

EM parameters of any EM physics list may be accessed or modified at initialization of Geant4 using:

- ♦ **G4EmCalculator**: easy access to cross sections and stopping powers (**TestEm0**)
- ♦ **G4EmParameters**: EM options alternative to UI commands
- ♦ **G4EmSaturation**: Birks effect (saturation of response of sensitive detectors)
- ♦ **G4ElectronIonPair**: sampling of ionization clusters in gaseous or silicon detectors
- ♦ **G4EmConfigurator**: add models per energy range and geometry region
- ♦ **G4NIELCalculator**: Helper class allowing computation of NIEL at a step, which should be added in user stepping actions or sensitive detector (**TestEm1**)

EM parameters of any EM physics list may be accessed or modified at initialization of Geant4 using:

- ◆ Use C++ interface to the **G4EmCalculator** (after physics list is initialized)
- ◆ Example for retrieving the total cross section of a process with name procName, for particle and material matName

```
#include "G4EmCalculator.hh"

{
    G4EmCalculator emCalculator;
    G4Material* material =
        G4NistManager::Instance()->FindOrBuildMaterial(matName);
    G4double density = material->GetDensity();
    G4double massSigma = emCalculator.ComputeCrossSectionPerVolume
        (energy, particle, procName, material)/density;
    G4cout << G4BestUnit(massSigma, "Surface/Mass") << G4endl;
}
```

Summary

- ◆ Overview of the main EM physics and classes in Geant4.
- ◆ There is a lot more physics in Geant4 (DNA project, etc, see also EM 2)
- ◆ Choosing the correct EM physics list while optimizing the simulation efficiency is the one of the most important issue you will face.
- ◆ We suggest starting with the standard list and if accuracy is extremely important to you, compare with option 4. This will give you an idea of your needs.